Positional constraints, sequence uniqueness, and stroke numerals in Indus seal inscriptions from Mohenjo-Daro: a statistical analysis

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Abstract

This study analyses a controlled subcorpus of 179 digitised Mohenjo-Daro unicorn seal inscriptions from the Indus Valley civilisation (c. 2600–1900 BCE) and presents evidence that these inscriptions exhibit the statistical properties of structured registration codes rather than phonetic writing, formulaic phrases, or arbitrary symbols. The restriction to a single site and animal type is methodologically deliberate: registration-code properties—sequence uniqueness, positional format, embedded numerals—can only be meaningfully tested within a homogeneous administrative context before cross-site generalisation is attempted. Five findings emerge. First, 98.3% of sign sequences are unique ($p < 0.001$ by permutation test), exceeding the rate expected under random assignment and ruling out formulaic repetition. Second, positional entropy profiles show low entropy at edges and high entropy in middle positions ($\chi^2 = 14.7$, $p = 0.005$), a pattern consistent with formatted codes (fixed classifiers framing variable identifiers). Third, 15.4% of signs are stroke-numerals concentrated in penultimate positions, indicating a mixed alphanumeric format. Fourth, the digit "2" constitutes 66.9% of numerals ($\chi^2 = 198.3$, $p < 0.001$ against uniform), excluding natural-quantity encoding. Fifth, eight seal pairs differ by exactly one sign, with variation restricted to middle positions (never edges). Twenty-eight prefix-sharing groups exhibit shared two-sign headers with divergent endings. These patterns are compatible with Mukhopadhyay's (2023) administrative interpretation: registration codes are the structural format that taxation, licensing, and commodity control require. Whether the findings generalise beyond Mohenjo-Daro unicorn seals to the broader Indus corpus is an open empirical question that the present study frames but does not answer.

Keywords: Indus script, Mohenjo-Daro, unicorn seals, registration codes, computational epigraphy, positional entropy, structured identifiers, non-linguistic coding

Contents

1 Introduction

The Indus script has resisted decipherment for over a century. Scholars have proposed readings in Sanskrit (Rao 1982), Proto-Dravidian (Parpola 1994), and other language families, while others have argued that the signs do not encode language at all (Farmer et al. 2004). Most recently, Mukhopadhyay (2023) demonstrated that Indus inscriptions encode administrative information—taxation, trade licensing, commodity control, and access control—rather than personal names or linguistic text. A conference on the Harappan script hosted by the Indian Ministry of Culture (New Delhi, September 2025; "Harappan Script—The Unsolved Mystery") underscored the field's continued uncertainty.

This paper examines whether a specific, well-controlled subset of the corpus—179 unicorn seal inscriptions from Mohenjo-Daro—exhibits the statistical properties of a structured registration system. The restriction to a single site, a single animal type, and a single object class is not a limitation to be apologised for but a *methodological requirement*. Registration-code properties (sequence uniqueness, positional format constraints, embedded numerals) can only be meaningfully tested within a homogeneous administrative context. Mixing sites, animal types, and object classes in an initial analysis would confound administrative variation with format variation: if Harappa and Mohenjo-Daro used different prefix conventions (as preliminary evidence suggests), a merged analysis would obscure precisely the positional structure we seek to identify. The correct sequence is: establish the format within one context first, then test whether it generalises.

The term "registration code" is used broadly and deliberately. The codes may have identified individuals, households, guilds, commodity classes, tax categories, administrative licences, territorial units, or any combination thereof. The statistical analysis establishes the *format*—a structured, mixed alphanumeric system with positionally constrained fields and near-total uniqueness. The *referent*—what the codes identified—is an archaeological question that the present statistical methods cannot answer and that this paper does not claim to resolve.

This proposal builds on Hunt's (2025a) argument that the Indus script operates as an information compression system, developed in Hunt (2025b) as a reconstruction of Indus social organisation. It is substantially compatible with Mukhopadhyay's (2023) administrative interpretation: taxation, licensing, and commodity control all *require* registration codes. The present study provides the quantitative analysis that transforms these positions into a statistically testable hypothesis.

If the structural findings presented here are confirmed on the broader corpus, the implications extend beyond epigraphy. No other known Bronze Age civilisation administered itself through abstract, non-linguistic, positionally constrained codes. Egypt and Mesopotamia recorded administration in phonetic writing—transcriptions of spoken language. The Indus system, as characterised here, operated at a higher level of abstraction: structured information encoding without linguistic mediation. This would make the Indus civilisation the earliest known society to have developed what may be called an *information architecture*—a system of structured, language-independent codes designed for administrative coordination across a linguistically diverse territory. The present paper does not claim to prove this characterisation; it presents the statistical evidence from one controlled subcorpus and identifies the tests required for confirmation.

2 Background and related work

2.1 The corpus

Approximately 4,000 inscribed objects have been recovered, predominantly steatite stamp seals with inscriptions averaging 4–5 signs (Mahadevan 1977). Over 400 distinct signs have been catalogued (Parpola 1994; Wells 2015).

2.2 Phonetic approaches

Rao (1982) proposed a Sanskrit reading. Parpola (1994) linked 425 signs to Proto-Dravidian via the rebus principle. Mahadevan (2014) proposed "Merchant of the City" for a recurring four-sign sequence.

2.3 Non-linguistic approaches

Farmer et al. (2004) argued the signs constitute non-linguistic symbols. Rao et al. (2009) showed the script's conditional entropy is intermediate between random and linguistic systems. Yadav et al. (2010) found well-defined text beginners and enders with strong bigram correlations using *n*-gram Markov analysis on the EBUDS corpus. Pollard (2025) proposed a merchant-mark system.

2.4 Administrative interpretation: Mukhopadhyay (2019, 2023)

Mukhopadhyay (2019) established nine functional sign-classes and formulaic phrase structures within Indus inscriptions. In a subsequent study, Mukhopadhyay (2023)—published in this journal—demonstrated that inscriptions encode taxation, trade licensing, commodity control, and access control. Crucially, Mukhopadhyay argued that the inscriptions *did not* encode personal names (anthroponyms), showing that longer inscriptions contain the same terminal and initial signs as shorter ones, with only core-informational portions increasing in length.

The present study agrees with Mukhopadhyay's central finding that the inscriptions encode administrative information rather than personal names or linguistic text. However, we propose that the format itself—structured codes with positionally fixed fields and near-total uniqueness—constitutes a registration system, whether the registered entities are tax categories, commodity licences, guild memberships, or (as we do not exclude but do not insist upon) individuals. Mukhopadhyay's taxonomy of sign-classes (taxation signs, commodity signs, metrological signs) is compatible with the registration-code format: each class occupies a specific positional field within the code structure. The format is the finding; the referent is an open question.

2.5 Information-theoretic framework

Hunt (2025a) argued in *Position Brief: Rethinking the Indus Script – Beyond Phonetic Assumptions* that the phonetic assumption may be the primary obstacle to understanding. Hunt (2025b) developed this in *Without Kings or Conquests: The Indus Script Deciphered and a Civilization Reconstructed*, connecting the script's structure to Indus social organisation through Living Information Theory (Hunt 2026).

3 Methods

3.1 Data source

The digitised corpus was obtained from an open-access GitHub repository ("indus-valley-script-corpus"), based on the Corpus of Indus Seals and Inscriptions (Parpola et al. 1987). The dataset contains 179 artefact sides from Mohenjo-Daro (M-1 through M-199), all unicorn seals (types I–V), with 1,003 total sign tokens and 182 unique sign types.

This corpus was selected over larger alternatives (e.g., the Interactive Corpus of Indus Texts, ICIT, with ∼5,500 entries) for a specific reason: it is the only Indus sign corpus that is fully open-access, machine-readable (JSON format with structured grapheme metadata), and independently downloadable without institutional access or administrator approval. Any researcher with Python 3 and an internet connection can reproduce every result in this paper within minutes. For a study whose central claim is statistical—and whose credibility therefore depends entirely on reproducibility—the choice of the most publicly accessible corpus is not incidental but foundational. Larger corpora with restricted access may yield stronger results, but those results would be unverifiable by the majority of the scholarly community. We prioritise transparency over sample size.

Limitation: This subcorpus represents approximately 3–4% of known inscriptions and is restricted to a single site and animal type. All claims are therefore qualified to this subcorpus; generalisation to the full corpus requires replication.

3.2 Analytical pipeline

Analyses were performed using Python 3 (standard library only):

1. Sign frequencies computed globally and per position.
2. Shannon entropy per position for fixed-length subsets ($n = 5, 6, 7$).
3. Sequence uniqueness with permutation test (10,000 iterations) against a null model preserving observed sign frequencies.
4. Bigram co-occurrence matrix and sparsity.
5. Stroke-sign identification from corpus metadata; P120 reclassified as composite.
6. Stroke-sign positional concentration per slot.

7. Digit frequency tested against uniform small-integer distribution (χ^2 test).
8. Hamming distance for all same-length pairs; significance tested against random-reassignment null model.
9. Prefix-sharing groups identified by shared first-2 and last-2 signs.
10. Combinatorial capacity estimation.

Scripts are provided as Supplementary Information and are reproducible from the public corpus.

3.3 AI-assisted computation

Statistical computations were performed using Claude Opus 4.6 (Anthropic, released 5 February 2026) as a computational tool. The AI retrieved the corpus from GitHub, wrote and executed the Python analysis scripts enumerated above, and returned numerical results. Consistent with the structure-first methodology (§4), the AI was restricted to operations in the constraint-extraction and null-model-testing stages: variable extraction, pattern detection, clustering, entropy computation, and statistical testing. The AI was not permitted to assign meaning, interpret sign functions, or generate hypotheses about referents—those operations were performed by the human authors, who are accountable for all interpretive and speculative claims. In accordance with Springer Nature policy, the AI is documented as a tool, not listed as an author.

Three technical capabilities of this model version, released six weeks before the analysis, were methodologically relevant: (a) a one-million-token context window enabling the entire corpus and all intermediate results to be held in working memory simultaneously; (b) the model identified the anomalous classification of P120 based on metadata inspection, a finding not anticipated by the human researcher; and (c) the model executed the full computational pipeline—repository cloning, JSON parsing, script writing, execution, and output interpretation—without manual coding intervention.

4 Conceptual framework

4.1 From symbolic identity to relational structure

The dominant analytical starting point in undeciphered-script studies has been to model sign systems as *symbolic identity mappings*: each sign carries an intrinsic meaning, and decipherment consists of recovering the mapping from sign to referent. This approach has shaped Indus script research, whether the proposed mapping is phonetic (sign $\rightarrow$ sound), logographic (sign $\rightarrow$ word), or taxonomic (sign $\rightarrow$ administrative category).

The present study takes a different approach, treating the inscriptions as a *constraint-governed system* in which meaning is not stored in individual symbols but emerges from their relational configuration—position, adjacency, and combinatorial admissibility. A sign in position 1 does not function as the same sign in position 4; not all sign pairs are admissible; and five constrained positions generate more structure than five unconstrained ones. This perspective draws on Hunt's (2025a, 2025b) argument that the Indus script operates as an information compression system, and on the broader principle—well established across information theory, linguistics, and archaeology—that structured systems are best characterised by their constraints before their contents are interpreted.

Definition 4.1 (Constraint-governed encoding system)**.** *A constraint-governed encoding system (i.e., one exhibiting statistically significant positional and combinatorial restrictions) is a symbolic architecture in which: (a) symbols occupy positionally defined roles (prefix, medial, suffix) that restrict their admissible values; (b) adjacency constraints govern which symbols may co-occur; (c) valid sequences are a small fraction of the combinatorial space, selected by the constraint set; (d) the system's information content resides in the constraint structure, not in individual symbol-referent mappings; (e) the system may function across linguistic boundaries because its logic is relational, not phonetic.*

The statistical analysis below tests whether the subcorpus exhibits the properties predicted by Definition 4.1—positional role differentiation, adjacency restriction, low sequence repetition, and mixed symbolic-numeric content—without presupposing what the codes *refer to*.

4.2 Analytical pipeline

The analysis proceeds through a structure-first sequence in which structural properties are established before interpretive labels are proposed. Raw corpus data are first *normalised* into standardised sign-ID sequences, with anomalous signs (e.g., P120) flagged and reclassified based on metadata. *Structural description* follows: positional frequencies, entropy profiles, bigram co-occurrence matrices, and stroke-sign distributions are computed—these describe what the system *does*, not what it *means*. Each finding then undergoes *null-model testing* (permutation, Friedman, χ^2, binomial); claims that do not survive rejection are discarded. Independent findings are checked for mutual *convergence*: a valid interpretation must be supported across multiple tests, not by any single finding alone. Among interpretations consistent with the converged evidence, the simplest is preferred (*parsimony*). Only then are *interpretive labels* assigned ("prefix classifier," "numeral field," "suffix"), explicitly marked as hypotheses (cf. Table 1: "Hypothesised role"). Finally, the selected hypothesis is treated as *provisional*: it will be revised or abandoned if specific predictions (§9.6) are disconfirmed.

4.3 Three evidential layers

To maintain clarity about the strength of different claims, all findings are explicitly assigned to one of three layers:

- **Empirical** (directly observed): positional regularities, co-occurrence patterns, uniqueness rates, entropy profiles. Reported in Results (§5).

- **Interpretive** (supported inference): positional role assignments, numeral identification, prefix-sharing groups. Discussed alongside empirical findings but marked as interpretive.
- **Speculative** (plausible but untested): civilisational implications, the lost-writing hypothesis, the guild-credential interpretation. Confined to later sections (§§7–8) and explicitly labelled.

5 Results

5.1 Sequence uniqueness

Of 179 inscriptions, 176 (98.3%) are unique sign sequences. A permutation test (10,000 iterations, preserving observed sign frequencies and inscription lengths but randomly reassigning signs to positions) produced a mean uniqueness of 91.2% (95% CI: 88.7–93.5%). The observed 98.3% exceeds this null ($p < 0.001$), indicating that the uniqueness is not an artefact of the sign frequency distribution but reflects intentional non-repetition.

5.2 Positional entropy profile

For 5-sign inscriptions ($n = 35$), Shannon entropy at each position (Table 1) reveals a low-high-low profile.

Table 1: Positional entropy for 5-sign inscriptions ($n = 35$). CI = bootstrap 95% confidence interval (10,000 resamples).

Position	Entropy (bits)	95% CI	Unique signs	Hypothesised role
1	3.13	2.6–3.5	15	Prefix classifier
2	4.44	4.0–4.7	25	Variable field
3	4.45	4.0–4.7	25	Variable field
4	3.77	3.3–4.1	20	Numeral field
5	3.98	3.5–4.3	23	Suffix classifier

A Friedman test across positions yields $\chi^2(4) = 14.7$, $p = 0.005$ (Kendall's $W = 0.105$, a moderate effect), confirming that entropy is not uniform across positions. The low-high-low pattern is consistent with a formatted code and inconsistent with both natural language (approximately uniform) and random symbols (uniformly high).

Overall unigram entropy: 6.3 bits; conditional entropy drops 62% when the preceding sign is known.

5.3 Stroke-sign numerals

Stroke signs exist in two parallel series: short (half-height, P121–P124) and tall (full-height, P144–P151). Combined, pure stroke signs constitute 154/1,003 tokens (15.4%). P120 was reclassified as composite (non-numeral) based on morphological metadata ("vertical + angled vertical").

5.4 Numeral positioning

Stroke signs concentrate in penultimate positions (Table 2).

Table 2: Stroke-sign percentage by position.

Position	5-sign ($n = 35$)	6-sign ($n = 32$)
1	5.7%	0.0%
2	17.1%	12.5%
3	20.0%	18.8%
4	**45.7%**	**40.6%**
5	5.7%	**37.5%**
6	—	6.2%

A χ^2 test of positional uniformity for stroke-sign placement in 5-sign inscriptions yields $\chi^2(4) = 27.4$, $p < 0.001$: numerals are significantly concentrated in penultimate positions.

5.5 Digit frequency distribution

The digit "2" constitutes 66.9% of all numeral occurrences (103/154). Tested against a uniform distribution over observed values (1–5), $\chi^2(4) = 198.3$, $p < 0.001$. The dominance of "2" excludes encoding of naturally distributed quantities and is consistent with ordinal classification (rank, category, licence tier).

Note: Benford's law, which applies to leading digits of multi-digit numbers, is not the appropriate null model for single-digit stroke counts. A uniform distribution is used as a conservative baseline. If numerals encoded ordinal ranks, one might expect a monotonically decreasing distribution (more "1"s than "2"s than "3"s); the observed dominance of "2" is anomalous under both uniform and decreasing models, which is precisely why it is informative.

5.6 Near-duplicates and prefix-sharing groups

Eight inscription pairs differ by exactly one sign (Hamming distance = 1). In all pairs, variation occurs at position 2 (5 pairs) or position 3 (3 pairs)—never at positions 1 or 5. Under a null model with random single-sign substitution (preserving position), the probability of all 8 differences avoiding edge positions is $p = 0.008$ (binomial test, edge probability = 0.4).

Twenty-eight prefix-sharing groups share identical first two signs with divergent endings. These may reflect administrative sub-categories (e.g., licences within a commodity class, units within a guild) or any other hierarchically nested registration structure. The term “prefix-sharing groups” is used instead of “family clusters” to avoid implying kinship without evidence.

5.7 Combinatorial capacity

A conservative 5-field estimate (10 prefixes $\times$ 30^3 medial signs $\times$ 5 suffixes) yields $\sim$1.35 million unique codes—exceeding Mohenjo-Daro’s estimated population ($\sim$40,000) and the civilisation’s estimated total (1–5 million).

6 Reconciliation of previous approaches

The registration-code interpretation does not require discarding previous scholarship:

Pictographic origin (widely accepted): correct. Registration systems routinely employ pictographic symbols (HS commodity codes, road signs, ISO standards).

Phonetic vocalisation (Parpola 1994; Mahadevan 2014): the codes were almost certainly read aloud, with phonetic realisation varying by language, region, and era. One reads “HS 8471.30” aloud without the vocalisation constituting a sentence.

Non-linguistic system (Farmer et al. 2004): confirmed and specified as structured registration codes with positionally constrained fields.

Merchant marks (Pollard 2025): compatible. Trade marks are a subset of registration codes.

Taxation and licensing (Mukhopadhyay 2019, 2023): substantially compatible. Mukhopadhyay’s sign-classes (taxation, commodity, metrological) correspond to positional fields within the registration format. The key difference: Mukhopadhyay identifies the *semantic content* of the fields; the present study identifies the *structural format* of the code. These are complementary findings.

Information compression (Hunt 2025a, 2025b, 2026): the registration system is a technology of information compression—encoding multiple administrative dimensions into $\sim$5 symbols with capacity exceeding the civilisation’s needs.

7 Implications if the registration-code hypothesis holds

A standardised registration system across five million square kilometres implies centralised format specification, uniqueness enforcement (98.3% in this subcorpus), and professional manufacture (mirror-engraved, kiln-fired steatite).

Many excavated seals show minimal wear, suggesting infrequent use—credentials for specific occasions, not daily tools. Seals themselves (not only clay impressions) have been found at Ur, Tell

Asmar, Bahrain, and other Mesopotamian sites, indicating that owners carried them when travelling abroad.

The foreign finds carry a crucial implication for the seal's function. An Indus administrative code—a tax category, a commodity licence, a municipal registration—would have had *no administrative validity in Mesopotamia*. Mesopotamian officials operated under Sumerian and Akkadian bureaucratic systems with their own cuneiform documentation. An Indus registration number would be meaningless in Ur, just as a Toronto municipal business licence is meaningless in Istanbul. Yet the seals travelled. The owner carried the seal not because a Mesopotamian port official could process its administrative content, but because the seal *identified the bearer as a member of a recognised trading entity*. The animal motif (unicorn, bull) was visually recognisable without reading the signs; the sign sequence confirmed membership to those within the Indus trading network. The seal functioned abroad as a *guild credential and guarantee of payment*—proof that the bearer represented a solvent trading house whose obligations would be honoured. This is precisely how medieval European merchant guilds operated: a guild seal guaranteed the bearer's transactions not because the foreign counterpart could read the seal's language, but because the guild's reputation stood behind it.

This narrows the functional interpretation. Domestically, the seal may have served multiple administrative purposes (taxation, licensing, access control, as Mukhopadhyay argues). Abroad, it served exactly one: proof of trading identity and creditworthiness. The fact that owners carried seals across thousands of kilometres of dangerous travel—rather than leaving them safely at home—indicates that the seal's value was *personal and reputational*, not merely bureaucratic. One does not risk losing an administrative form number in a foreign port. One carries a credential that opens doors.

India's dominant funerary tradition of cremation (Kenoyer 1998) explains the scarcity of seal deposits and the absence of perishable writing: steatite shatters and organic materials are destroyed. The $\sim$4,000 surviving seals represent a fraction of those that existed.

The absence of monumental architecture, royal tombs, and military imagery in the Indus civilisation has long puzzled archaeologists. A civilisation that organises itself through registration rather than royal decree would appear exactly thus: egalitarian in visible expression, sophisticated in invisible infrastructure.

8 Modern structural parallels

The registration-code hypothesis gains force when the Indus seal system is compared with modern registration technologies. The parallels are not metaphorical; they are structural.

Vehicle registration plates. A license plate is a formatted code (prefix = jurisdiction, alphanumeric middle = unique identifier, sometimes a suffix = vehicle class). It is physically affixed to the object it identifies. It is manufactured by authorised facilities, not by the owner. It is readable at a glance by anyone who knows the format, regardless of language. It is designed for durability: stamped or embossed in metal so that the code survives even if paint decays—the raised characters remain legible through corrosion, dirt, and weathering. The Indus seal is carved in relief (not painted or inked) and fired to convert soft steatite into hard enstatite: the code survives even if surface treatment is lost. Both systems embed the information in the *physical topology* of the object, not in a superficial layer.

Passport numbers. A passport number is a structured alphanumeric code (country prefix + type code + serial number + check digit). It identifies a person but encodes no linguistic content—"AB 1234567" does not mean anything in any language. It is read aloud when needed, with vocalisation varying by the speaker's language ("Alpha Bravo One Two Three..." in English; entirely different phonemes in Hindi or Finnish). The document is manufactured centrally, carried personally, presented at borders, and—in many cultures—kept among one's most valued possessions. Indus seals were manufactured by specialists, carried on the person, presented at checkpoints, found in Mesopotamia (carried abroad), and placed in graves.

Company registration numbers. A business entity is assigned a unique code by a central authority (Companies House number in the UK, EIN in the US, CIN in India). The code identifies the entity across all administrative interactions: taxation, contracts, customs, legal proceedings. The Indus civilisation, with its standardised weights, uniform urban planning, and long-distance trade networks, would have needed precisely such a system for guilds, trading houses, or commodity producers. Mukhopadhyay's (2023) evidence that inscriptions encode taxation and licensing information is directly compatible: tax codes *are* registration codes.

Commodity codes (HS system). The Harmonised System assigns 6-digit codes to every traded commodity worldwide (e.g., HS 0901.11 = unroasted, not decaffeinated coffee). The code has a hierarchical structure: chapter (2 digits) $\rightarrow$ heading (4 digits) $\rightarrow$ subheading (6 digits). It is used on shipping documents, customs forms, and trade agreements. If the Indus seals encoded commodity classes (as Mukhopadhyay argues), a hierarchical code—animal motif (broad class) + prefix (sub-class) + numeral (grade or quantity unit) + suffix (administrative status)—would be the natural format.

Roman administrative abbreviations. A useful comparison can be drawn with Roman administrative inscriptions (e.g., IMP CAES M AVR ANTONINVS AVG P P COS III), which exhibit ordered structural fields: title (IMP), category (CAES), personal identifier (M AVR ANTONINVS), status (AVG P P), and ordinal marker (COS III). Roman numerals in this context function as ordinal or rank indicators rather than raw counts—COS III denotes the third consulship, not three consuls—providing a useful comparison for interpreting the non-uniform distribution of numerals in the Indus corpus. The written structure remained stable across regions despite variation in spoken language. This comparison does not imply historical connection but illustrates how administrative systems operating under similar constraints can converge on comparable structural solutions: ordered fields, mixed alphanumeric content, ordinal numerals, and durable manufacture.

Common design principles. All these systems share features observed in the Indus seals:

- Structured format with positionally fixed fields.
- Mixed alphanumeric content (categorical symbols + serial numbers).
- Near-total uniqueness within the active population.
- Central manufacture by authorised agents, not self-produced.
- Physical durability: information embedded in the object's topology, not in a perishable surface layer.
- Portability: carried by or affixed to the registered entity.
- Language-independence: the code functions across linguistic boundaries.
- Phonetic readability without linguistic content.

No modern system exhibits *all* of these features simultaneously. The Indus seal system does. This convergence of design principles across 4,500 years suggests not cultural continuity but functional convergence: when a civilisation needs to register entities across a large territory with diverse languages, the solution space is narrow, and the resulting format is predictable.

8.1 Phonetic vocalisation drifts; the code does not

A powerful modern analogy clarifies why Parpola's phonetic readings may be genuine observations without the script being phonetic writing. The NATO phonetic alphabet reads the letter "B" as "Bravo." Before 1956, the same letter was vocalised as "Baker" (Joint Army/Navy), "Beer" (RAF 1942), or "Benjamin" (French military). The letter has not changed; its spoken realisation has drifted across decades, institutions, and languages.

Chinese: the decisive case. The Chinese writing system provides the strongest existing demonstration that a sign system can function across mutually unintelligible phonetic traditions. The character for "water" is read as *shuǐ* in Mandarin, *séui* in Cantonese, *chúi* in Hakka, *sui* in Japanese *on'yomi*, and *mizu* in Japanese *kun'yomi*. The character is identical; the phonetic realisations are entirely different words in mutually unintelligible languages. A Cantonese speaker and a Mandarin speaker cannot understand each other's speech, yet they read the same document. The script encodes meaning, not sound. If the Indus civilisation spanned five million square kilometres with diverse linguistic communities—as archaeological and genetic evidence suggests—a non-phonetic coding system would be not merely possible but *necessary* for inter-regional communication.

The @ symbol: phonetic chaos within one generation. The @ sign, a single globally standardised symbol, is vocalised as "at" in English, *Klammeraffe* ("spider monkey") in German, *chiocciola* ("snail") in Italian, *sobaka* ("dog") in Russian, *strudel* in Hebrew, *kukac* ("worm") in Hungarian, and *arroba* in Spanish. These names were assigned independently within a single generation (the 1990s, as email spread). No two cultures vocalise the symbol identically. All use it identically. This demonstrates that a symbol adopted across linguistic boundaries will *inevitably* develop divergent phonetic traditions—and that the divergence carries no information about the symbol's function.

Pictographic origins: convergent or shared. The pictographic origin of Indus signs is beyond dispute—many visibly depict fish, trees, humans, jars. But pictographic origin does not determine a system's function. The Phoenician *aleph* (ox head), Greek *alpha*, and Etruscan A all derive from a pictogram of an ox. The pictographic origin is shared; the functions diverged radically—from a consonantal marker (Phoenician) to a vowel (Greek) to a borrowed alphabetic sign (Etruscan). Pictographic ancestry tells us where signs *came from*, not what they *do*. The Indus "fish" sign may have originated as a picture of a fish. Whether it then encoded the sound *mīn* (Parpola), a commodity class (Mukhopadhyay), or a field in a registration code (present hypothesis) cannot be determined from pictographic origin alone.

Sign diversity points to logographic origin. The $\sim$400 sign inventory is too large for an alphabet (typically 20–30 signs) and too large for a syllabary (typically 40–100). It is, however, the range expected of a logographic or logo-syllabic system—comparable to early Sumerian cuneiform ($\sim$600 in the Uruk IV period) and Egyptian hieroglyphs ($\sim$700 in the Old Kingdom). This suggests that Indus signs originated as logograms: each representing a word, concept, or category. Over time, some signs may have acquired phonetic values through the rebus principle (as happened in Sumerian, Egyptian, and Chinese)—but the registration codes on the seals need not have used those phonetic values. The seals may represent a *frozen logographic layer* of a larger writing system whose phonetic dimension operated on perishable media and is now lost.

The lost writing: a speculative but testable hypothesis. The following argument is explicitly speculative and is offered as a hypothesis, not a conclusion. A civilisation with standardised weights, uniform urban planning, long-distance trade, and sewerage *may* have possessed full writing beyond what survives on the seals—but this does not follow necessarily. The Inca Empire maintained sophisticated administration across a vast territory using quipu (knotted cords) without developing a writing system in the conventional sense, demonstrating that administrative complexity does not require literacy. Whether the Indus civilisation resembled literate Mesopotamia or non-literate Inca in this respect is an open question.

If perishable writing did exist, it would have been on the materials used throughout South and Southeast Asian history: palm leaves, birch bark, cotton cloth, bamboo strips, wooden boards. None of these survive India's monsoon climate, termite activity, and funerary cremation over 4,000 years.

The absence of long Indus texts is not evidence of a non-literate civilisation. It is evidence of a *cremating civilisation in a tropical climate*. Every ancient South Asian literary tradition—Vedic, Buddhist, Jain—was transmitted orally for centuries before being committed to writing, and even then the earliest surviving manuscripts date to the first centuries CE, some 2,000 years after the Indus period. If the Vedas survived only through oral transmission until writing was used centuries later, an Indus literary tradition could have existed and been lost completely—leaving only the stone registration codes as inadvertent survivors.

The $\sim$400 sign inventory supports this: it is too rich for a pure registration system (which would need 50–100 signs) but appropriate for a logographic writing system of which the seals preserve only the administrative subset. The codes drew from a larger inventory also used for full writing—but the full writing perished with its media.

Phonetic instability without established literacy. In civilisations with proven phonetic writing—Egypt, Mesopotamia, Greece—the written form stabilised pronunciation to some degree: literate speakers converge toward the written standard. In a civilisation *without* proven phonetic writing, there is no stabilising mechanism. Spoken language drifts at the rate of oral transmission: within a single generation, across a single city, between social classes. The idea that a consistent phonetic reading can be recovered from a civilisation that left no phonetic texts—spanning 700 years across a multilingual territory—requires the assumption that pronunciation was stable without a stabilising mechanism. The comparative evidence (Chinese, @, NATO alphabet) argues precisely against this.

Applied to the Indus context: a seal bearing P324–P122–P385 could have been vocalised as Proto-Dravidian words in the 26th century BCE, as substrate language words in the 23rd century BCE, and as Akkadian syllables by a Mesopotamian port official. All vocalisations would be "correct." Parpola's readings may capture one such tradition—real, historically grounded, yet not evidence that the script encodes Dravidian.

This explains why phonetic approaches yield plausible but unverifiable readings. The readings are plausible because vocalisation traditions existed. They are unverifiable because the code admits *multiple* traditions, none canonical. Searching for the "correct" phonetic reading of the Indus script is like searching for the "correct" name for @.

8.2 Manufacturing process as evidence for central registration

The physical production of an Indus seal constrains the hypothesis space. The process involved multiple specialised steps:

1. **Material preparation.** Raw steatite (soapstone, Mohs hardness 1–2) was cut into standardised squares of 2–3 cm—dimensions consistent across the civilisation (Kenoyer 1998).
2. **Intaglio carving of the animal motif.** The animal was carved in negative relief (recessed), requiring skilled sculptors capable of rendering anatomical detail (musculature, skin folds, horn curvature) in miniature and in mirror image.
3. **Intaglio carving of the sign sequence.** Each sign was carved in mirror image so that the impression would read correctly. This is the critical step: mirror writing of arbitrary symbol sequences requires either (a) memorisation of each sign's reversed form, or (b) use of a template or guide. Either implies specialised training.
4. **Boss or perforated knob on the reverse.** A loop or protrusion was formed for cord suspension, indicating the seal was designed to be worn.
5. **Firing.** The carved seal was heated to 900–1100°C, converting steatite (talc) to enstatite—a hard, white, glazed ceramic. This transformation is irreversible: errors cannot be corrected post-firing. The code, once fired, is permanent.

Several implications follow:

Self-production is excluded. Steps 2–4 require professional skills (mirror carving, kiln access, material knowledge). A merchant, farmer, or administrator could not produce their own seal, just as a citizen cannot print their own passport. Seal production was necessarily controlled by specialists.

Error intolerance implies verification. Firing is irreversible. A single miscarved sign renders the seal unusable. The engraver must verify the sequence before committing to fire. If the sequence was assigned by a registry, the engraver would work from an authoritative record. If self-selected, the error cost falls on the customer—but the near-total uniqueness (98.3%) suggests systematic assignment, not self-selection (which would produce accidental duplicates).

Workshop concentration implies institutional control. Archaeological evidence from Harappa indicates that seal production was concentrated in specific workshop areas (Kenoyer 1998), not distributed across households. Concentrated production is consistent with regulated manufacture under administrative oversight.

The fired seal is a tamper-proof document. Once enstatite, the seal cannot be recarved (hardness 5–6, vs. steatite's 1–2). The code is permanently fixed. Alteration would require destroying the seal and commissioning a new one—detectable if a registry exists. This is functionally equivalent to modern tamper-evident security features on identity documents.

The manufacturing process thus implies: professional engravers, centralised workshops, pre-production verification of the code sequence, and post-production tamper resistance. At the level of a single city (Mohenjo-Daro, population $\sim$40,000), this constitutes a *registration bureau*—an institutional body that assigns codes, authorises production, and maintains records. Whether such coordination extended across the entire civilisation or operated city-by-city is an open question, but the format standardisation across sites suggests at minimum a shared protocol, and possibly a shared authority.

8.3 The seal as a professional credential: why it was buried, not reused

A persistent puzzle in Indus archaeology is the find context of seals: some in workshops and streets (administrative use), some in houses (personal storage), and some in the rare inhumation graves (funerary deposit). The registration-code hypothesis, combined with the manufacturing evidence, suggests a unified explanation.

If the seal was an *institutionally issued professional or status credential*—analogous to a modern medical licence, bar association card, military commission, or guild membership certificate—then several otherwise puzzling features become coherent:

Non-transferability. A professional credential is issued to a specific entity (person, guild chapter, trading house) and is not valid in other hands. When the bearer dies, the credential is void. It cannot be inherited, resold, or reassigned—just as a physician's licence does not pass to their children. This explains why seals are found in graves: the credential accompanies the bearer because it *has no further use*. It identifies only one entity, and that entity is gone.

Minimal wear. Professional credentials are *shown*, not *used* as tools. A lawyer shows their bar card; they do not stamp documents with it. A military officer shows their commission; they do not press it into clay daily. The near-pristine condition of many Indus seals—sharp edges, crisp engraving, no abrasion—is precisely what one expects of a document that is presented on demand but not physically applied repeatedly. The seal was proof of registration, not a production instrument.

Not buried as a personal keepsake. The alternative—that seals were buried as sentimental objects or amulets—does not explain the standardised format. People do not bury their taxi licence out of sentiment; they bury it because it *is* them, institutionally. The seal was the bearer's place in the administrative order. Without the bearer, the seal is meaningless; with the bearer, the seal is authority. Burying it together is not affection—it is closure.

Institutional recall vs. grave deposit. If the seal was institutional property (like a government-issued ID card), one might expect it to be returned to the issuing authority upon the bearer's death for cancellation and reuse of the code. The fact that some seals are found in graves rather than returned suggests either: (a) the code was permanently retired upon the bearer's death (as modern social security numbers are), making the physical object disposable; or (b) the seal was considered the bearer's property despite being institutionally issued (as modern university diplomas are). Both scenarios are consistent with the registration-code hypothesis.

The absence of duplicate seals for the same code. If seals were merely commercial stamps, one would expect backup copies or replacements for worn-out originals. The near-total sequence uniqueness (98.3%) and the absence of same-code duplicate seals across excavated sites suggest a *one-seal-per-code* rule. This is precisely how credentials work: there is one original, and its loss or destruction is administratively significant.

9 Discussion

9.1 What the constraint analysis reveals

The five statistical findings—sequence uniqueness, positional entropy profile, numeral concentration, digit-frequency anomaly, and edge-avoidant near-duplicates—converge on a single structural characterisation: the Mohenjo-Daro unicorn seal inscriptions exhibit the properties of a constraint-governed encoding system (Definition 4.1). Positional roles are differentiated (low-entropy edges, high-entropy middle). Adjacency constraints are strong (only 1.66% of possible bigrams are realised). Valid sequences are a small, intentionally non-repeating subset of the combinatorial space. And the system mixes symbolic and numeric elements in positionally segregated fields.

This characterisation is richer than "registration code," which implies a flat symbol-to-referent mapping. The constraint-architecture framing preserves what the data actually show: that meaning in this system is *relational*—it arises from position, adjacency, combinatorial admissibility, and context, not from individual signs in isolation. A sign in position 1 does not function as the same sign in position 4. A stroke-numeral adjacent to a prefix classifier does not carry the same structural role as the same numeral adjacent to a suffix. The system's information content resides in the constraint structure, not in a look-up table of sign meanings. While the present analysis does not assign specific semantic domains, the structural roles identified are consistent with systems that encode not only identity but also classification and context-dependent state information.

9.2 What the analysis does not determine

The constraint analysis establishes the *format* of the encoding system. It does not—and cannot—determine the *referent*: whether the codes identified individuals, guilds, commodity classes, tax categories, territorial units, or some other administratively relevant entity. This is not a limitation of the method; it is a principled boundary. Referent identification requires archaeological context (find-spot, associated artefacts, cross-site comparison with metadata) that is absent from the digitised subcorpus analysed here.

9.3 Relationship to Mukhopadhyay's semantic analysis

Mukhopadhyay (2023) identified what the positional fields *encode*: taxation signs, commodity signs, metrological modes. The present study identifies *how* those fields are structurally organised: as a constraint-governed system with quantifiable positional entropy, measurable uniqueness, and testable format properties. These are complementary findings. Mukhopadhyay's semantic categories map onto our structural positions: her "taxation signs" may correspond to our prefix classifiers; her "commodity signs" to our high-entropy medial field; her "metrological modes" to our numeral field. Testing this correspondence on the full ICIT corpus is the most important next step.

A methodological distinction should be made explicit. Unlike approaches that assign specific semantic values to individual signs or sign-classes—however carefully reasoned (e.g., Mukhopadhyay 2019, 2023; Parpola 1994)—the present study operates entirely at the level of structural syntax: positional constraints, entropy profiles, combinatorial properties, and null-model comparisons. No sign in this paper is assigned a meaning, a phonetic value, or a semantic domain. This deliberate abstinence carries a cost (the results say less about content) but also a benefit: it avoids the risk of premature semantic commitment—the assignment of linguistic or categorical meanings that, once embedded in the analytical framework, become self-reinforcing and difficult to falsify. By treating the inscriptions as abstract information-bearing structures whose content remains undetermined, the analysis establishes formal properties that any future semantic interpretation must be consistent with, without prejudging what that interpretation will be.

9.4 Continuity of sign forms: Tamil Nadu graffiti and Indus signs

A recent morphological study by Rajan and Sivanantham (2025), commissioned by the Tamil Nadu State Department of Archaeology, digitised over 15,000 graffiti-bearing potsherds from 140 archaeological sites across Tamil Nadu and found that approximately 60% of 42 base signs and their variants have morphological parallels with Indus script signs, with 90% of all graffiti marks showing some degree of similarity. The chronological gap is substantial: the Indus civilisation occupied the Copper Age (c. 2600–1900 BCE), while the Tamil Nadu sites span the Iron Age (c. 800–300 BCE), with at least a millennium separating them.

The constraint-governed encoding hypothesis offers a natural explanation for this continuity. If the Indus signs functioned as *administrative codes*—commodity markers, category classifiers, registration elements—rather than as phonetic writing, then their persistence on ceramics across a thousand years and two thousand kilometres follows from the persistence of the *administrative function*, not the persistence of a language. A sign denoting a commodity class on a sealed storage jar in Mohenjo-Daro in 2200 BCE and a morphologically similar sign scratched on a potsherd in Keeladi in 600 BCE need not encode the same word in the same language; they may encode the same *category* in the same administrative tradition. Commodity marks, property marks, and quality stamps are among the most conservative symbolic systems known: potters' marks in the Mediterranean persisted across language changes for millennia.

This interpretation strengthens rather than weakens the case for cultural continuity between the Indus civilisation and later South Indian traditions. A purely phonetic reading of the Indus script would require demonstrating that the *same language* was spoken across the gap—a claim that is inherently difficult to verify without bilingual texts. The administrative-code interpretation requires only that the *same marking practices* persisted—a far more parsimonious claim, directly supported by the morphological evidence that Rajan and Sivanantham have documented.

Moreover, the Indus seals' constraint-governed format implies administrative sophistication (centralised registration, standardised formats, professional manufacture, uniqueness enforcement) that speaks to the organisational capacity of the originating civilisation. If the Tamil Nadu graffiti preserve traces of this tradition, then South Indian Iron Age communities inherited not merely signs but a system of civic organisation. This does not diminish the achievement; it elevates it. The Indus civilisation did not merely write—it *administered*, across five million square kilometres, using a technology of structured information that persisted for over a thousand years.

9.5 The world's first known information civilisation

The claim in the title requires explicit justification. We argue that the Indus civilisation, as characterised by the present analysis, represents the earliest known case of a society administering itself through *abstract, non-linguistic, structured information codes*—and that no comparable system is attested in any other Bronze Age civilisation.

Comparison with contemporary civilisations. Sumerian cuneiform and Egyptian hieroglyphs are writing systems: they encode spoken language. Administrative records in Mesopotamia (e.g., Ur III tablets) and Egypt (e.g., tax rolls, census records) are transcriptions of sentences—subject, verb, object, quantities in natural language. They are readable because they are linguistic.

The Indus system, as characterised here, is not linguistic. It is a *positionally constrained, combinatorially unique, mixed-symbolic-numeric code* with no demonstrated phonetic content. Its closest modern analogues are not other ancient scripts but modern administrative technologies: structured databases, postal code systems, commodity classification schemes (HS codes), vehicle registration systems. These are information architectures—systems that encode, store, and retrieve structured data without linguistic mediation.

What makes this "first." Other ancient civilisations used non-linguistic marking systems (potters' marks, ownership stamps, tally marks). But none, to our knowledge, developed a system with *all* of the following properties simultaneously: (a) standardised positional format across a territory of five million square kilometres; (b) near-total sequence uniqueness ($>$98%) implying systematic non-repetition; (c) mixed symbolic-numeric content in positionally segregated fields; (d) professional centralised manufacture with tamper-resistant firing; (e) combinatorial capacity exceeding the population by orders of magnitude; (f) persistence of sign forms for over a thousand years (Rajan and Sivanantham 2025).

Qualification. The claim is "first *known*," not "first." Earlier or contemporary systems may have existed on perishable media and been lost. The claim applies to what the archaeological record currently attests. It is also provisional: if the full ICIT corpus analysis reveals that the constraint structure does not generalise beyond Mohenjo-Daro unicorn seals, the claim must be withdrawn. But if it does generalise—across sites, animal types, and object classes—then the Indus civilisation stands as the earliest documented case of a society whose administrative backbone was not writing, but information architecture.

9.6 Falsifiability

The hypothesis makes the following testable predictions, any of which, if falsified, would substantially weaken or refute it:

1. *Cross-site replication.* If the positional entropy profile and uniqueness rate do not replicate on Harappa unicorn seals, the hypothesis is site-specific, not system-wide.
2. *Cross-animal comparison.* If bull, elephant, and rhinoceros seals exhibit the same sign vocabulary (not merely the same format), the animal motif does not function as a categorical field.
3. *Longer inscriptions.* If inscriptions of 10+ signs exhibit the same entropy profile as 5-sign inscriptions (rather than showing format extension with additional fields), the pattern may reflect linguistic syntax rather than constrained encoding.
4. *Tablet vs. seal divergence.* If tablet inscriptions exhibit statistically identical properties to seal inscriptions, the stamping-linked function is not supported.
5. *Linguistic null.* If short administrative phrases in Sumerian Ur III texts with comparable vocabulary size produce a statistically indistinguishable entropy profile, the pattern is not diagnostic of non-linguistic encoding specifically.
6. *Tamil Nadu graffiti.* If the morphologically similar signs documented by Rajan and Sivanantham (2025) occupy the same positional slots and exhibit the same constraint structure as

their Indus counterparts, this would support administrative continuity. If they do not, the morphological similarity may reflect convergent geometry rather than functional inheritance.

The hypothesis is provisionally accepted and will be revised or abandoned if any of the above tests produces disconfirming evidence.

10 Limitations

1. The subcorpus (179 artefacts) represents 3–4% of known inscriptions. All claims apply to Mohenjo-Daro unicorn seals; generalisation requires replication on the full ICIT corpus.
2. No cross-animal or cross-site analysis. The statistical profile may differ for bull, elephant, or other seal types.
3. Absence of find-spot metadata limits archaeological contextualisation.
4. Stroke-sign identification follows Parpola's numbering; some assignments are uncertain.
5. Entropy estimates for 5-sign inscriptions ($n = 35$) have non-negligible confidence intervals.
6. The registration-code hypothesis specifies the format but cannot determine the referent (individual, guild, commodity, licence) from corpus statistics alone.

11 Future directions

Priority extensions: replication on the full ICIT corpus (∼5,500 texts) with animal-type and site metadata; testing whether sign vocabularies differ across animal types; geographic variation analysis; comparison with Mukhopadhyay's (2023) sign-class taxonomy to determine whether positional fields correspond to her functional categories; and formal comparison with modern registration coding systems as a statistical benchmark.

Data availability

The corpus is available at `https://github.com/mayig/indus-valley-script-corpus` (CC-BY-4.0). Python scripts are provided as Supplementary Information.

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approach as “a welcome exception to the papers now poured on the decipherment of the Indus script,” while maintaining his commitment to phonetic decipherment. This generous engagement with an alternative framework exemplifies the scholarly openness that the field requires, and the present paper’s reconciliation framework (§6) is offered in the same spirit: not to displace phonetic research, but to show how it captures a genuine layer of a multi-layered system.

Author contributions

B.K. conceived the structured-identifier hypothesis, designed and executed the statistical analysis pipeline (with AI-assisted computation), produced all quantitative results, and wrote the manuscript. T.A.H. developed the constraint-governed analytical framework and the structure-first interpretive methodology applied in §4; provided the information-theoretic foundation; contributed to the interpretation of statistical results; and critically revised the manuscript. The empirical findings (§**??**) are independent of the theoretical framework and stand on their own statistical merits. Both authors approved the final version.

Artificial intelligence usage

Claude Opus 4.6 (Anthropic, 5 February 2026) was used as a computational tool for corpus retrieval, script execution, and statistical computation, as described in the Methods section. All hypotheses, interpretations, and claims are the responsibility of the human authors.

Competing interests

The authors are aware that in January 2025 the Government of Tamil Nadu announced a prize of US$1 million for the decipherment of the Indus script (the Iravatham Mahadevan Prize). The present study may be considered relevant to that prize. The research was initiated independently of and prior to the prize announcement, and the authors’ conclusions are not influenced by its existence.

Ethical approval

This article does not contain any studies with human participants performed by any of the authors. The research analyses publicly available digitised archaeological data.

Informed consent

This article does not contain any studies with human participants performed by any of the authors.

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Figure legends

No figures are included in this manuscript. All results are presented in Tables 1–2.